

# Traffic engg in computer networks

## module-1 1m

1. what is a network protocol
2. what is Data rate or bandwidth
3. what is jitter
4. give atleast 2 reasons for packet loss in general
5. what are ping and traceroute tools
6. what is transmission and propagation delays
7. what terms do you need to calculate propagation delay over a wire
8. what terms do you need to calculate transmission delay over a wire
9. explain the terms : attenuation, noise, interference
10. how to use iperf3 in linux to calculate the network performance metrics between 2 machines

## module-1 2.5m

1. list all the protocols in OSI layer
2. what is digital data transmissison preferred over analog transmission
3. difference between amplifier and repeater. What does repeater better than amplifier for digital transmission
4. explain switched ethernet, and how does it build the MAC address tables within the LAN
5. what is the problem with sending a broadcast message in a LAN, how does spanning tree protocol solve this problem
6. given size of a frame is 100 bytes, what should be the datarate/bandwidth of the sender so that transmission delay equals propagation delay. Length of wire is 50km, propagation speed is  $2 \times 10^8$  m/s.

## module-1 5m

1. What is a NAT device in ipv4. What problem does it solve and how?
2. differences between tcp and udp protocol
3. (a) explain the terms : subnet, network address, subnet mask  
(b) list all the ip addresses in the subnet 192.168.20.73/3. explain with proper reason  
(c) given that 192.168.20.73 lies in a subnet with subnet mask 255.255.255.240. will 192.168.20.81 lie in the same subnet?. Explain with reason
4. what is the size of the smallest possible subnet that can give seperate subnets for 4 divisions containing 60, 62, 12,120 devices respectively
5. Suppose a 128-Kbps point-to-point link is set up between Earth and a rover on Mars. The distance from Earth to Mars (when they are closest together) is approximately 55 Gm, and data travels over the link at the speed of light— $3 \times 10^8$  m/sec.  
(a) Calculate the minimum RTT for the link.  
(b) Calculate the delay  $\times$  bandwidth product for the link.  
(c) A camera on the rover takes pictures of its surroundings and sends these to Earth. How quickly after a picture is taken can it reach Mission Control on Earth? Assume that each image is 5 MB in size.
6. Calculate the latency (from first bit sent to last bit received) for:  
(a) A 1-Gbps Ethernet with a single store-and-forward switch in the path, and a packet size of 5,000 bits. Assume that each link introduces a propagation delay of 10  $\mu$ s and that the switch begins

retransmitting immediately after it has finished receiving the packet

(b) Same as (a) but with three switches.

(c) Same as (b) but assume the switch implements cut-through switching: it is able to begin retransmitting the packet after the first 128 bits have been received.